C PROGRAMMING • LECTURE 02

Variables, Data Types & Input / Output

scanf • printf • format specifiers • typecasting

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Batch • CSE-60

```
int age = 20;
float price = 99.5f;
char grade = 'A';

printf("%d", age);
```

What is a Variable?

A variable is a named memory location used to store a value.

- Think of a variable as a labeled box in memory.
- The value can change while the program runs.
- Every variable has a name and a data type.
- Example: age, marks, salary, grade.

```
int age = 20;
float marks = 85.5f;
char grade = 'A';
```

Declaring & Initializing Variables

```
int age;
age = 20;

int marks = 95;
```

Declaration

Tell the compiler the variable's type and name.

Initialization

Give a variable its first value.

Use meaningful names such as totalMarks when possible.

Basic C Data Types

The data type tells C what kind of value a variable stores.

int

Whole numbers

10, -5, 100

float

Decimal values

3.14, 9.5

double

More precision

3.14159265

char

One character

'A', '7', '#'

The % symbol connects data to formatting

Common beginner specifiers.

int

`printf: %d`

`scanf: %d`

float

`printf: %f`

`scanf: %f`

double

`printf: %f`

`scanf: %lf`

char

`printf: %c`

`scanf: %c`

Important: double uses %lf with scanf, while printf uses %f for double.

printf()

printf prints formatted output.

```
int age = 20;
printf("Age = %d\n", age);

float x = 5.5f;
printf("x = %.2f\n", x);
```

Key ideas

- Format string
- Value to print
- \n → new line
- %.2f → 2 digits after decimal

scanf()

scanf reads input according to a format string and stores it in variables.

```
int age;
printf("Enter age: ");
scanf("%d", &age);

printf("Age = %d", age);
```

Why &age?

scanf needs the address of the variable so it can store the input there.

Remember:

```
scanf("%d", &age);
```

Reading More Than One Value

```
int a, b;
scanf("%d %d", &a, &b);
printf("Sum = %d", a + b);
```

Input example

```
10 25
```

Both values are read in order.

Explicit Type Conversion

Typecasting converts a value to another type.

```
int a = 5, b = 2;

float x = (float)a / b;
printf("%.2f", x);
```

Key idea

(float)a tells C to treat a as a float in that expression.

Result → 2.50

A Common Beginner Mistake

```
int a = 5, b = 2;
float x = a / b;

printf("%.1f", x);
```

Result: 2.0

a / b is integer division first.

Fix:

```
float x = (float)a / b;
```

Arithmetic with Variables

```
int a = 10, b = 3;

a + b   // 13
a - b   // 7
a * b   // 30
a / b   // 3
a % b   // 1
```

Remember

- / with integers gives integer division
- % gives the remainder
- Use float/double for fractional results

char vs String

char = ONE character

```
char grade = 'A';
printf("%c", grade);
```

String = sequence of characters

```
char dept[30] = "CSE";
printf("%s", dept);
```

C uses character arrays to represent strings.

Type Conversion

Implicit

C automatically converts types in many expressions.

```
int a = 5;
double x = a;
```

Explicit / Cast

You request the conversion yourself.

```
int a = 5;
double x = (double)a;
```

Things to Watch Out For

Wrong specifier

```
printf("%d", 3.14);
```

Missing &

```
scanf("%d", age);
```

Integer division

```
5 / 2 → 2
```

Wrong char quotes

```
char c = "A";
```

Uninitialized variable

```
int x; printf("%d", x);
```

Easy Practice

Write a program for each problem.

1. Read two integers and print sum, difference and product.

2. Read radius and print circle area. Use π = 3.14159.

3. Read name, age and marks, then print them neatly.

4. Read a character and print its ASCII value.

Medium Practice

Use variables, scanf, printf and typecasting.

1. Convert total seconds into hours, minutes and seconds.
2. Read two integers and print their average as a floating-point value.
3. Calculate simple interest from principal, rate and time.
4. Convert Celsius to Fahrenheit.
5. Read 3 subject marks and print total + average to 2 decimals.

Find the Output

Try first. Then compile and verify.

```
int a = 7, b = 2;
float x = a / b;
float y = (float)a / b;

printf("%.1f\n", x);
printf("%.1f\n", y);
```

Question

What will be printed?

Explain WHY the two values are different.

Build a Student Result Program

- Input student name, ID and marks of 3 subjects.
- Calculate total and average.
- Print all information in a clean format.
- Make sure the average is floating-point, not integer division.

Expected skills

Variables
scanf()
printf()
Format specifiers
Typecasting

What You Should Know Today

- Variable = named storage for a value.
- Basic types: int, float, double, char.
- printf() → output; scanf() → input.
- Format specifiers connect data to formatted I/O.
- Use & with ordinary scalar variables in scanf.
- Typecasting explicitly converts a value.
- Watch out for integer division.

NEXT STEP

Practice before the next lecture.

LECTURE 02 COMPLETE

Write code.
Make mistakes.
Understand why.

Then solve the tasks without looking at the solution.